Part 1 – Report

Data Cleaning

Step 1: Cleaning duplicated rows.

Removed 80 duplicated rows from the dataset (2002 to 1922).

Step 2: Column Removal

I found several columns with a significant percentage of missing data. Since those columns would probably offer little helpful information, I chose to eliminate them when the ratio of missing values surpassed 80%.

- **Note**: There are columns with only a few distinct values—such as Campaign Record Type, Campaign Categories and Campaign Subtype. So, these could be encoded using techniques like One Hot Encoding. In this case, they were left as-is to maintain readability.

Step 3: Standardizing the Format of Full Names

The "Full Name" column had several different formats, so I standardized it. This guarantees consistency and makes analysis easier.

Step 4: Formatting Campaign Name:

The Campaign Name column contained two extra elements. First, it had a value in square brackets [] indicating the Campaign Subtype, which was removed because this information is already stored in another column. Second, it included the year in parentheses. While the year could be useful and extracted into a new column "Year," I decided to remove it entirely because all campaigns in the dataset occur between September and October 2024. In future datasets with a wider range of years, extracting this information could be relevant.

Step 5: Formatting Placeholder or Missing Values

I avoided deleting rows, even though some fields contained missing data, because other important information such as Name or Project Name was present. For better readability, I also reformatted certain fields where missing or placeholder values appeared as "-", "ninguna" (in Primary Affiliation: Account Name), or were left blank, replacing them with "None" or "Unknown."

- **Recommendation:** This line if these fields are intended for visual dashboards or AI/ML models. In that case, the entries will remain blank, allowing proper detection of missing values (NA).

- **Note:** We noticed some cases where Registered is 0 while Attended is 1. We kept the data as is, assuming that a person can register on-site, which the database does not capture.

API Integration:

The cleaned dataset was enhanced using a public API and a small Python script. In this instance, the "Full Name" field was used to determine the likely gender of the individuals using the Genderize.io API (with a >98% accuracy). To provide more context for analysis, the recovered data was combined with the dataset.

- **Recommendation**: Genderize.io has a usage limit of 100 requests per day, so it is important to be cautious with its usage. Additionally, the API integration was purely demonstrative and intended to illustrate how external data can be combined with the cleaned dataset. This approach can be adapted to other public APIs

Data Structuring:

For effective administration and analysis, the cleaned dataset can be kept in a relational database. Each participant should have a unique identifier (id) as an integer primary key. Campaign dates, including start_date and end_date, should be stored as DATE fields. Textual information such as campaign_record_type, campaign_categories, campaign_name, campaign_subtype, full_name, account_name, email, and gender should be stored as VARCHAR fields, with appropriate maximum lengths (e.g., 100–255 characters depending on the field). Engagement data, such as registered, attended, and receives_newsletter, should be stored as BOOLEAN fields to indicate true/false status.

Also, if we plan to integrate with CRM, such as Salesforce, each participant could be represented as a Lead or Contact, with campaigns connected by a Campaign object. It is possible to store categorical fields such as custom fields, such as Campaign Record Type, Campaign Categories, Campaign Subtype, and Gender. By establishing distinct tables for Participants and Campaigns and managing the many-to-many relationships between them with a junction table, the database could be normalized to eliminate redundancy.

Repository Link:

https://github.com/AlejandroHancco/JobSample-Propel/tree/develop/Part1_Data_Cleaning